Q.13) Difference Between Regression and Classification

Ans:
Regression and Classification are two major types of **Supervised Machine Learning** techniques.

- **Regression** is used to predict a **continuous numerical value**.
- **Classification** is used to predict a **categorical/class label**.

Difference Between Regression and Classification

Regression	Classification
Predicts a **continuous numerical value**.	Predicts a **discrete class or category**.
Produces a **numerical output**.	Produces a **class/label output**.
The output is **continuous**.	The output is **categorical/discrete**.
Finds the relationship between input variables and a **continuous target**.	Assigns data to one or more **predefined classes**.
Examples: **House price, salary, temperature, sales prediction**.	Examples: **Spam/not spam, disease/no disease, pass/fail**.
Common algorithms: **Linear Regression, Multiple Linear Regression, Polynomial Regression**.	Common algorithms: **Logistic Regression, Decision Tree, KNN, SVM, Naive Bayes**.
Evaluation metrics include **MAE, MSE, RMSE, R^2 Score**.	Evaluation metrics include **Accuracy, Precision, Recall, F1-Score, Confusion Matrix**.
Graphical output is often represented using a **line or curve**.	Graphical output is often represented using **decision boundaries/classes**.
Example output: **₹5,50,000 house price**.	Example output: **Spam or Not Spam**.

Example

1. Regression

If we want to predict the **price of a house** based on its area, number of rooms, and location:

Input → ML Model → ₹5,50,000

The output is a **continuous numerical value**, so it is a **Regression problem**.

2. Classification

If we want to determine whether an email is **Spam or Not Spam**:

Input → ML Model → Spam

The output is a **category/class**, so it is a **Classification problem**.

In Short

Regression → Continuous Numerical Output

Classification → Categorical Output

Conclusion

Both Regression and Classification learn from **labeled training data**, but they mainly differ in the type of output they produce. **Regression predicts numerical values, whereas Classification predicts categories or classes.**